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(See 6 8)

MILITARY SPECIFICATION

COMPASS, CYRO, MINIATURE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1 1 Scope This specification covers the requirements for a miniature gyro compass to furnish true heading indications

2 APPLICABLE DOCUMENTS

2 1 Government documents

2 1 1 Specifications and standard The following specifications and standard form a part of this specification to the extent specified herein Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H I (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for
MIL-C-5015 - Connectors, Electrical, Circular Threaded, General Specification for
DOD-B-15072 - Battery Storage Lead-Acid, Portable Type 12V-15P - L/D - A or B (Metric)
MIL-A-15303 - Audible Signals Alarms, Bells, Buzzers, Horns and Sirens, Electronic, Shipboard
MIL-E-16400 - Electronic Interior Communication and Navigation Equipment, Naval Ship and Shore General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MILITARY (Continued)

MIL-M-17413 - Motors, Direct Current, Integral H P , Naval Shipboard
MIL-E-17555 - Electronic and Electrical Equipment Accessories, and Provisioned Items (Repair Parts) Packaging of
MIL-S-20708 - Synchros, Control Differential Transmitter, Type 26V - 11CD x 43

STANDARD

MILITARY

MIL-STD-710 - Synchros, 60 and 400 Hertz

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity)

2 3 Order of precedence In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3 REQUIREMENTS

3 1 First article When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4 4 and 6 3)

3 2 Description The miniature gyro compass equipment shall consist of the following

(a) Compass unit
(b) Control unit (may be integral with the compass unit)
(c) Speed unit
(d Alarm unit (for single or multiple compass installation (see 6 2 1))
(e) Visual alarm (see 6 2 1)
(f) Isolation transformer
(g) Power supply unit (see 6 2 1)

3 3 General features In addition to the requirements specified herein the equipment shall be in accordance with MIL-E-16400

3 4 Material All parts except those where strength or hardness is essential, shall be constructed of nonferrous, nonmagnetic materials.

3 4 1 Recovered materials Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the

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intended use The term "recovered materials" means materials which have been collected or recovered form solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified

3 5 Construction The miniature gyro compass equipment shall operate from a 3-phase, 115-volt, 400-hertz (Hz) supply with tolerances specified in MIL-E-16400 Total 400-Hz power requirements to operate the compass equipment excluding the power supply unit shall not exceed 400 volt amperes Cable entrance into enclosures shall be either by stuffing tubes or AN type connectors in accordance with MIL-C-5015 Enclosures shall be dripproof The compass equipment shall not be damaged by failure of power to the equipment Components shall pass the vital equipment vibration requirements of MIL-E-16400 as specified herein Components shall pass the type A high shock requirements of MIL-S-901 for lightweight equipment as specified herein Internal resilient mounts may be used in the compass unit or in an integral compass and control unit only to meet the shock requirement External resilient mounts shall not be used

3 5 1 Compass unit The compass unit shall, by means of a gravity controlled gyroscope responsive to earth rates, continuously seek and indicate the true meridian The true heading of the carrier (a vehicle or ship) shall be readable from an integral dial of the compass unit This unit shall be provided with a protective enclosure constructed for three-point deck or shelf mounting Mounting holes shall provide for plus or minus 5 degrees adjustment A fore and aft line so marked shall be engraved on the base of the enclosure to facilitate alignment to the carrier's centerline The fore and aft alignment marks shall be referred to the innermost heading reference alignment mark within plus or minus 6 minutes of arc The compass unit shall perform as a gravity controlled gyroscope at any latitude between 75 degrees south and 75 degrees north At latitudes over 75 degrees to the poles, the compass unit shall perform as a dependable direction indicating gyro with a drift rate not in excess of 0 06 degree per hour

3 5 1 1 Magnetic field The compass unit shall indicate true heading with an error not to exceed plus or minus 1 5 degrees when the unit is subjected to a magnetic field of 10 gausses

3 5 1 2 Maximum rate of turn The follow-up system of the compass unit shall turn in azimuth at any rate up to 100 degrees a second The follow-up shall be fitted with alarm contacts which will indicate failure of the follow-up system by means of an indicator lamp on the control unit The compass equipment shall not be susceptible to damage upon failure of the follow-up system (see 4 7 9)

3 5 1 3 Settling Means shall be provided for electrically leveling and slewing the gyroscopic element of the compass unit so that it can be leveled and set within 10 degrees of the true heading. A tilt indicator shall be provided either on the compass unit or on the control unit The compass unit shall meet the settling test specified in 4 7.1.

3.5 1 4 Noise The gyro compass equipment shall withstand the tests for airborne audible, structure-borne and radio frequency noise specified in 4 7 13

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3 5 1 5 Bearings. The gyro rotor or rotors shall be constructed for a minimum life expectancy of 15,000 hours

3 5 1 6 Rotor vibration The gyro rotor or rotors shall operate from a zero speed to 15 percent overspeed without any critical periods of vibration in the operating range or any objectionable critical periods throughout the entire range

3 5 1 7 Roll and pitch clearance The compass unit shall be operable under conditions of a roll of 45 degrees or a pitch of 45 degrees on each side of the vertical

3 5 1 8 Speed and course The effect of the carrier's speed and course on the true heading indication shall be automatically compensated for the gyroscopic element to indicate the true meridian instead of the dynamic meridian Effects of change of latitude on the true heading indication shall be compensated for by the manual introduction of latitude signals in the control unit

3 5 1 9 Pitch acceleration The compass unit shall be not adversely affected by pitch accelerations up to 1,000 degrees per second and pitch velocities up to 50 degrees per second In addition, it shall be not adversely affected by a fore and aft horizontal acceleration of 1 gravity (g) of 2 seconds duration or by a momentary fore and aft horizontal acceleration of 3 g's

3 5 1 10 Heading transmission The compass unit shall transmit heading at 1 and 36 speed The synchro transmitters shall be 18TR4 standard Navy synchros in accordance with MIL-S-20708 and MIL-STD-710

3 5 1 11 Adjustment protection Provision shall be made for preventing unauthorized changing of the compass unit adjustments

3 5 1 12 Size The size of the compass unit with or without and integral control unit shall not exceed a nominal 2 cubic feet

3 5 2 Control unit The control unit shall contain the amplifiers, computer circuitry, power supplies, meters alarm indicator lights and switches to control and monitor compass operation The enclosure for the control unit unless made integral with the compass unit shall be constructed for three-point bulkhead mounting The volume of the control unit shall not exceed a nominal 1-1/4 cubic feet

3 5 2 1 Accessibility The power supplies and amplifiers shall be removable for examination and testing with plug and jack connections Test cables 10 feet long fitted with proper connectors to allow the removable subassembly to be electrically connected to the control unit shall be included unless the front cover of the control unit is arranged to serve as a service platform In the latter case, the plug half of the connector may be fitted with a shorter electrical cable which will allow servicing the subassemblies on the front cover.

3 5 2 2 Operating knobs Operating knobs on the control panel shall be constructed or protected so that accidental contact will not disturb them or change their setting Introduction of latitude compensation signals shall be by a knob and scale on this unit

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3 5 2.3 Instruction plates. Instruction plates shall be placed on the control unit where necessary.

3 5 2 4 Electron tubes and solid state semi-conductor devices For power rectification, solid state semi-conductor devices shall be used instead of electron tubes

3 5 3 Speed unit The speed unit shall, to facilitate making the speed compensations specified in 3 5 1 8, accept a 60-Hz, 0 to 40 knot synchro input where 9 degrees of synchro rotation equal 1 knot The speed unit shall generate a signal proportional to speed as required by the design of the compass equipment This unit shall be in an enclosure suited for three-point bulkhead mounting

3 5 3 1 Speed input Provision shall be made to manually insert the speed input using a dial calibrated 0 to 40 knots

3 5 3 2 Size The size of the speed unit shall not exceed a nominal 1/8 cubic foot

3 5 4 Alarm unit The alarm unit shall consist of necessary relays and components to actuate one or more audible alarms like the B5DSF4 bell of MIL-A-15303 and one or more visual alarms in the event of failure in a synchro signal amplifier, which is used with, but is not a part of, the compass equipment, or when certain critical circuits or functions of the compass equipment become inoperative, operate under par, or when there is a power failure For a multiple compass installation the alarm unit shall contain the necessary components to in addition to the above, actuate the audible or visual alarms upon closure of an external alarm circuit in any one of the compass systems installed

3 5 4 1 Size The size of the alarm unit shall not exceed a nominal 1/6 cubic foot This unit shall be constructed for three-point bulkhead mounting

3 5 5 Visual alarm The visual alarm shall contain one or more lights A steady light shall indicate normal functioning of the compass, power supply or synchro signal amplifier, a flashing light shall indicate an alarm condition in a compass, in a synchro signal amplifier or failure of the power supply

3 5 5 1 Size The visual alarm shall be not larger than a nominal 1/6 cubic foot The alarm shall be constructed for three-point bulkhead mounting

3 5.6 Isolation transformer The isolation transformer shall be provided for interposing between the carrier's 115 volt, 3-phase, 400 Hz supply and the compass equipment

3.5.7 Power supply unit A static type power supply unit and appropriate controls shall be furnished to automatically provide 400-Hz power to the compass and loads on the compass transmitters to keep the compass and those loads operating normally during periods of ship's supply failure or intolerably low voltage or low frequency in the ship's supply The unit and its controls shall be constructed so that for ship's voltage and frequency values below the lower limits in MIL-E-16400 the unit will operate from a 120 volt direct current (Vdc)

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source (type BB255/U, BB252/U, or BB256/U batteries of DOD-B-15072 or a rectifier) The unit shall be automatically switched to charge the batteries upon restoration of normal ship's power The unit shall be suited for continuous operation when operating from ship's supply or from a direct current (dc) supply (When used with a rectifier the power supply unit shall not furnish power to or through the rectifier to other loads on the rectifier) The unit shall not be damaged if connected for an indefinite period of time to partially or fully discharged batteries and the compass equipment shall not be damaged when dependent on the power supply unit during this time Protection against damage from discharged batteries shall not be dependent on fuses or circuit breakers The unit in addition to supplying the power required to operate the compass equipment shall supply an additional 250-watt, 0 8 power factor (p f), 1-phase 400-Hz external equipment load The total harmonic content of the output of the unit shall be not more than 5 percent of the fundamental and the maximum of any one harmonic shall not be more than 2 percent of the fundamental The unit shall be not less than 60 percent efficient when operating from the ship's supply and not less than 70 percent efficient when operating from a dc supply Voltage and frequency regulation when the unit is operating from dc in the range 105 to 120 volts shall be plus or minus 2 volts and plus or minus 2 Hz for no load to full load The unit shall provide a battery charging voltage of 129 ± 2 volts when operating from any ship's supply voltage and frequency within the limits of MIL-E-16400 Facility shall be built into the unit wherewith to set the battery charging current rate within the following limits for the specific type of battery being used of those listed above

	Charging rates in amperes	
Battery type	Start	Finish
BB255/U	1 5	1
BB252/U	8	4
BB256/U	14	7

The power supply unit shall be constructed so that for line-to-line faults or 3-phase line-to-line faults for a duration of at least 5 seconds no fuses or circuit breakers shall blow or open, no damage shall be done to the unit and upon clearing the fault the unit shall continue to operate as specified The size of the power supply unit shall not exceed 5 cubic feet The unit shall be constructed for bulkhead mounting and shall be constructed for a minimum of 5000 hours mean time between failures

3 5 8 Interchangeability Equipment shall be constructed so that any compass unit of the same mark and modification will operate as specified in this specification with any control and amplifier unit or any speed transmitting unit of the corresponding mark and modification without requiring any modification other than minor adjustment of electrical circuits such as are practicable aboard ship In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

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3 5 9 Temperature and humidity The compass equipment shall operate within the accuracy limits specified in 3 5.10 at any temperature from 40 to 149 degrees Fahrenheit (°F) when tested (see 4 7 3) If necessary, an integral heater or blower may be provided to accomplish this requirement The equipment shall be protected for failure of the heater or blower At temperatures from 40 to 110°F the turntable error specified shall be allowed (see 3 5 10)

3 5 10 1 Extended storage The equipment shall meet the extended storage requirements of MIL-E-16400

3 5 10 Allowable error The limits of allowable errors for the gyro compass equipment shall be as shown in table I for the latitude of the contractor's plant or that of the commercial laboratory and are referred to the 1-speed synchro transmitter The limits include synchro transmitter errors settling point error and allowable alignment errors.

TABLE I Limits of allowable errors

Name of test	Allowable errors (± degrees)
Settling point (difference)	0 2
Settling point	0 75
Supply line voltage and frequency variation	1 0
Turntable	0 75
Small angle Scorsby	1 0
Large angle Scorsby	2 0
Effect of magnetic fields	1 5
Correction for speed and latitude	0 15

3 5 11 Service record book The contractor shall record the following data in the Service Record Book, NAVSHIPS 708, which is Government-furnished

Total test time in hours compass has sustained _____
Date compass was accepted for delivery _____
Damping _____ percent
Compass period _____ minutes
Average settling point during settling tests _____
Serial number of gyro assembly _____
Settling time _____
Test procedure for shipboard personnel to recheck the above data

3 5 12 Drawings When specified in the contract or order, drawings shall be prepared (see 6 2.2).

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1.1 Responsibility for compliance All items must meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material

4.1.2 Inspection system When specified in the contract or order an inspection system program plan shall be prepared (see 6.2.2)

4.2 Classification of inspections The inspection requirements specified herein are classified as follows

(a) First article inspection (see 4.4)
(b) Quality conformance inspection (see 4.5)

4.3 General procedures The procedures, definitions and conditions applicable for inspection to be performed on the equipment shall be as specified hereinafter Except where specifically indicated all tests shall be performed at room temperature (nominally 75°F) with normal supply voltage and frequency with the speed and latitude settings which prevail and with the compass operating in its normal mode of operation for latitudes 0 to 75 degrees All readings shall be recorded in degrees, tenths of a degree and to the nearest hundredth of a degree Unless otherwise specified, readings of compass indications for all tests shall be taken at 15-minute intervals Furthermore, for each test the compass shall be aligned to a reference heading which shall be accurate to within plus or minus 30 seconds The compass shall be aligned in azimuth to the reference heading within plus or minus 3 minutes The compass error shall be measured either by reading out the 36-speed compass transmitter outputs on a 36-speed dial of a gyro repeater or recorder and comparing the reading with the Scorsby reading, or by using a synchro control transformer geared at 36-speed to the Scorsby test stand, furnishing signals from the 36-speed compass transmitters to the control transformers and reading the control transformer rotor voltage on a calibrated meter or recorder Unless specific test headings are specified, the selected headings shall have the concurrence of the Government representative The compass binnacle or output transmitters of the compass shall not be shifted to center any settling point error

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4 3 1 Equipment mounting for inspection requirements

4 3.1 1 Equipment The compass equipment shall be tested on a MK5 or MK6 or equivalent Scorsby test stand

4 3 1 2 Mounting The compass unit shall be mounted on the Scorsby test plate, or on the turntable, so that the fore and aft line of the compass enclosure base lies in the vertical plane of the fore and aft line of the turntable, or Scorsby test plate The fore and aft line of the Scorsby test plate shall be so oriented that the gimbal axes of the gyro compass will always remain parallel to the corresponding axes of the Scorsby test stand

4 3 1 3 Adjustment After the compass unit has been mounted on the Scorsby test plate or turntable, no change shall be made affecting the position of the compass unit with respect to the Scorsby test plate or turntable, or in the lubber line of the compass unit, to correct for change in the settling point of the compass unit

4 4 First article inspection First article inspection shall consist of the examination and tests specified in table II

TABLE II First article inspection

Examination and tests	Requirement	Method
General examination	MIL-E-16400	4 6
Enclosure	3 5	MIL-E-16400
Settling point	3 5 1 3 and 3 5 10	4 7 1
Damping	3 5 11	4 7 4
Turntable	3 5 10	4 7 6
Maximum rate of turn	3 5 1 2	4 7 17
Supply line voltage and frequency variation	3 5 10	4 7 2
Angular freedom	3 5 1 7	4 7.18
Small angle Scorsby test	3 5 10	4 7 7
Interrupted power test	3 5 7	4 7 8
Power transfer test	3 5 7	4 7 8.1
Follow-up amplifier failure test	3 5 1 2	4 7 9
Alarm unit test	3 5 4	4 7 19
Large angle Scorsby test	3 5.10	4 7 10
Correction for speed and latitude test	3 5 1 8 and 3 5 10	4 7 11
Directional gyro test	3 5.1	4.7 5
Noise	3.5 1 4 and MIL-E-16400	4 7 13
Effect of magnetic field	3.5.1.1	4.7.15
Vibration	3.5	4 7 16
Temperature and humidity	3.5.9	4 7 3
Rotor vibration test	3 5 1 6	4 7 12
Dielectric strength	MIL-E-16400	4 7 14 1
Insulation resistance	MIL-E-16400	4 7.14 2
Shock	3 5	4.7 16

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4 5 Quality conformance inspection. All equipment shall be subjected to the examination and tests listed in table III which shall be performed in the order listed

TABLE III Quality conformance inspection

Examination and tests	Requirement	Method
General examination	MIL-E-16400	4 6
Dielectric strength	MIL-E-16400	4 7 14 1
Insulation resistance	MIL-E-16400	4 7 14 2
Settling point	3 5 1 3 and 3 5 10	4 7 1
Damping	3 5 11	4 7 4
Turntable	3 5 10	4 7 6
Small angle Scorsby test	3 5 10	4 7 7
Interrupted power test	3 5 7	4 7 8
Power transfer test	3 5 7	4 7 8 1
Follow-up amplifier failure test	3 5 1 2	4 7 9
Alarm unit test	3 5 4	4 7 19
Large angle Scorsby test	3 5 10	4 7 10
Correction for speed and latitude test	3 5 1 8 and 3 5 10	4 7 11
Directional gyro test	3 5 1	4 7 5

4 6 General examination The general examination shall be conducted as specified in MIL-E-16400 No operating test is required as part of the general examination

4 7 Test methods

4 7 1 Settling The gyro compass shall settle with an error not greater than the settling point error of 3 5 10 on any heading according to the conditions of 6 4 6 and the following

4 7 1 1 Readings Settling readings shall be taken at 5-minute intervals near the peaks and at 1-minute intervals on the peaks whenever the compass unit is required to settle in preparation for any part of the test During the settling period the compass unit shall be kept stationary (except for the dynamic settling test) and vertical on the Scorsby test stand and the correction devices shall be adjusted for zero speed at the local latitude setting

4 7 1 2 Waiver of readings At the beginning of any part of the test if the last readings for the preceding part show that the compass unit has settled as defined in 6 4 6, the settling readings may be waived for the part of the test being started, unless settling readings are requested by the contractor However, the settling readings shall not be continued beyond the time at which the compass unit has first settled as defined in 6 4 6, unless such settling readings are required for conducting the test

4 7 1.3 Rapid settling When preset to within 10 degrees of the true heading, the compass unit shall come to and indicate within 1.5 degrees of the true heading in less than 1 hour when started at a nominal temperature of 20°F with the compass unit stationary

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4 7 1 4 Dynamic settling. The compass shall settle in less than 4 hours with an error not greater than the settling point error of 3 5 10 when started up and run under large angle Scorsby test motions at a nominal room temperature of 75°F and when set within 10 degrees of the true heading

4 7 1 5 Time to settle Not more than 4 hours shall be allowed at any time for the compass unit to settle for any set of test conditions specified herein

4 7 2 Supply line voltage and frequency variation The gyro compass shall, after being settled as defined in 6 4 6 with voltage at 115 volts and frequency at 400 Hz, sustain a sudden change to the power supply voltage and frequency limits specified herein, and shall meet the accuracy requirements of 3 5 10 on any heading while undergoing a turntable test and on any heading while undergoing the small and large angle Scorsby tests Reading shall be taken every 15 minutes

(a) Frequency at 380 Hz, voltage at 109 volts, with the frequency not changing more than 2 Hz per hour after reaching 380 Hz
(b) Frequency at 420 Hz, voltage at 121 volts, with the frequency not changing more than 2 Hz per hour after reaching 420 Hz
(c) Frequency and voltage several times varying between the extremes of the power supply limits allowed by MIL-E-16400 with the variation occurring in some instances suddenly and in other instances more slowly A test on only one heading is adequate

4 7 3 Temperature and humidity test For this test the compass shall be settled according to 6 4 6 on one heading The following procedure shall apply with the compass operating in its normal mode The compass shall at no time exceed the turntable error of 3 5 10 at temperatures from 40 to 110°F and shall at no time exceed twice the turntable error of 3 5 10 for higher temperatures The power supply unit shall be tested at the extreme temperatures of this test after its temperature has stabilized to demonstrate that it will meet the requirements of 3 5 7

Hours	Ambient temperature and relative humidity during test	Angle of tilt
	With the compass at an equilibrium temperature of 40°F, start and settle the compass This shall require not more than 4 hours Record the time required	
0-21	40°F/95 percent relative humidity	Normal
21-27	Raise the temperature in 6 hours to 149°F, maintain 95 percent relative humidity	Normal
27-152	Temperature 149°F, relative humidity 95 percent at alternate 1 hour periods, that is, 1 hour at 95 percent relative humidity and the next hour with relative humidity uncontrolled Measure air temperature external and internal to the enclosure and check parts for hot spots, at the end of 125 hours.	Normal
152-156	Lower the temperature and humidity to room conditions, nominally 75°F and 50 to 60 percent relative humidity	Normal

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Throughout the test the compass error shall be recorded either continuously or at regular 15-minute intervals. Hot spot temperatures shall not exceed the maximum dictated by affected insulation as specified in MIL-E-16400 and shall be performed according to MIL-E-16400 The maximum internal temperature rise shall be recorded and shall not exceed 90°F from the ambient Equipment installation and the air change rate capability of the test chamber shall be in accordance with MIL-E-16400

4 7 3 1 Extended storage For this test the procedure shall be in accordance with MIL-E-16400 The equipment after stabilizing at the room temperature shall settle with an error not greater than the settling point error of 3 5 10

4 7 4 Damping test The compass unit shall be settled as defined in 6 4 6, on any heading designated by the contractor, with concurrence of cognizant Government representative The sensitive element shall then be carefully precessed 5 degrees from the established meridian and leveled so that the level reading is in such direction as to increase the deflection from the established meridian The compass unit shall then be allowed to settle as defined in 6 4 6 The time required to settle shall be not more than 4 hours

4 7 4 1 Mean percentage of damping The mean percentage of damping shall be calculated as follows

$$\frac{(a) - (c)}{(a) + (b)} \times 100$$

where

a = Amplitude first peak
b = Amplitude second peak
c = Amplitude third peak

4 7 4 2 Allowable variation The mean percentage of damping shall not differ more than plus or minus 10 percent from the value given by the contractor in accordance with 3 5 11 and shall be between 60 and 80 percent

4 7 5 Directional gyro test The compass unit shall be settled as defined in 6 4 6 on any heading designated by the contractor with the concurrence of the cognizant Government representative The compass shall then be operated in the directional gyro mode for a 2-hour period The drift rate for this test shall not exceed 0 6 degree per hour

4 7 6 Turntable test The compass unit shall be mounted on a Scorsby test stand and settled as defined in 6 4 6 It shall then be given a 4-hour test on each of four headings of the Scorsby test stand as designated by the contractor with the concurrence of cognizant Government representative. The compass unit error shall at no time during this test exceed the limit specified in 3.5 10 for turntable.

4.7.7 Small angle Scorsby test The compass unit shall be mounted on a Scorsby test stand and shall be settled as defined in 6 4 6 It shall be given an 8-hour test on each of the following six headings

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(a) Any two cardinal headings (0, 90, 180 and 270 degrees)
(b) Any two intercardinal headings (45, 135, 225, and 315 degrees)
(c) Any two semi-intercardinal headings (22 5, 67.5, 112.5, 157.5, 202.5, 247 5, 292 5, and 337 5 degrees)

4 7 7 1 Sequence of test On each heading the sequence of test shall be as follows.

Time	Condition
4 hours	Roll, pitch and yaw
2 hours	Roll only
2 hours	Pitch only

4 7 7 2 Sequence of headings The contractor, with concurrence of the cognizant Government representative, shall designate the sequence of the headings which are to be used during this test The amplitude of roll on each side of the vertical shall be 15 degrees, with a period between 8 and 10 seconds The amplitude of pitch on each side of the vertical shall be 5 degrees, with a period between 6 and 8 seconds The amplitude of yaw on each side of the Scorsby test stand heading shall be 5 degrees with a period between 8 and 10 seconds

4 7 7 3 Allowable error The compass unit error shall not exceed the limit specified in 3 5 10 for this test

4 7 8 Interrupted power test This test shall be conducted as specified in 4 7 8 1 except that ship's power shall be interrupted instead of reduced gradually The allowable error shall be that specified in 3 5 10 for the large angle Scorsby test

4 7 8 1 Power transfer test The compass shall be settled as defined in 6 4 6 on any heading designated by the contractor with the concurrence of the cognizant Government representative The compass shall then be operated under large angle Scorsby conditions on one heading The normal power supply shall then be reduced to a voltage or frequency less than the lower ship's supply limits allowed by MIL-E-16400 Transfer of operation to the power supply unit shall be effected automatically The compass shall operate on the power supply unit for several minutes and then with ship's power restored to normal voltage and frequency shall transfer back to ship's power automatically The compass error as shown by readings taken for 1 hour or more beginning 5 minutes after transfer back to ship's power shall not exceed the allowable error specified in 3 5 10 for the large angle Scorsby test

4 7 9 Follow-up amplifier failure test The compass unit shall be settled as defined in 6 4 6 on any heading selected by the contractor with concurrence of the cognizant Government representative. Then operation of the amplifier in the follow-up system shall be interrupted for a period of 5 minutes, simulating a failure The Scorsby turntable shall then be operated under large angle conditions The amplifier shall then be reconnected and the compass unit shall function normally.

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4.7 10 Large angle Scorsby test This test shall be conducted as specified in 4 7 7 for the small angle Scorsby test, except that the amplitude of roll shall be 40 degrees on each side of the vertical, with a period between 8 and 10 seconds, the amplitude of pitch shall be 10 degrees on each side of the vertical, with a period between 6 and 8 seconds, and the amplitude of yaw shall be 10 degrees on each side of the Scorsby test stand heading with a period of 10 and 12 seconds The compass unit error shall not exceed the limit specified in 3 5 10

4 7 11 Correction for speed and latitude test

4 7 11 1 Procedure The compass unit shall be settled as defined in 6 4 6 or any heading designated by the contractor with concurrence of cognizant Government representative The speed and latitude correction shall be set for zero speed and local latitude, and the Scorsby turntable shall be rotated to a heading designated by the contractor with concurrence of cognizant Government representative The correction shall then be set for a specified speed and the compass allowed to settle The true heading indication of the compass unit shall be recorded for the zero speed input and the specific speed input

4 7 11 2 Repetition The procedure specified in 4 7 11 1 shall be repeated for several well-distributed combinations of speed and heading

4 7 11 3 Computed corrections The indicated corrections obtained shall be compared with the values computed by the following formula

$$\tan C = \frac{S \cos C^1}{e \cos L - S \sin C^1}$$

where

C = Virtual meridian correction
S = Speed setting in knots
C^1 = Course setting (headings)
e = 902 46 knots
L = Local latitude in degrees

4 7 11 4 Allowable error The difference between the indicated correction and the corresponding computed correction shall not exceed the allowable error specified in 3 5 10 for this test

4 7 11 5 Speed data For the correction for speed and latitude test, the compass equipment shall be operated from the ship's underwater log or from a device which simulates the output of the log equipment

4 7 12 Rotor vibration test. The gyro rotor or rotors shall be operated from zero speed to normal operating speed, and shall be operated as specified in 4 7.12 1 and 4 7 12.2 The rotor or rotors shall have no critical periods of vibration in the operating range, nor any objectionable critical periods throughout the entire range at any time during this test Temperature rise shall not exceed that specified in 4 7 12 1 and 4 7 12 2

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4 7 12 1 Operation in gas or vacuum Each gyro rotor, if designed for vacuum or gas, shall be run in its case at an overspeed of at least 15 percent for a period of not less than 1 hour, during which period there shall be no objectionable vibration or noise of the gyro rotor, and the temperature rise of the bearings shall not exceed 67(°C) above the ambient (room) temperature, as determined by the thermometer method specified in MIL-M-17413 or any other method satisfactory to the Government representative

4 7 12 2 Operation in air Where the construction does not involve hermetic sealing and requires gyro rotor or rotors to operate in a vacuum or gas the rotor or rotors shall operate in air at atmospheric pressure with a 40-percent reduction in speed for 4 hours without exceeding a temperature rise of 67°C above the ambient (room) temperature as determined by the thermometer method specified in MIL-M-17413 or any other method satisfactory to the Government representative

4 7 13 Noise Each unit of the compass equipment shall be tested separately for compliance with the radio frequency noise requirements of 3.5 1 4 and for compliance with the airborne audible and structureborne noise requirements of 3 5 1 4 for other than low-noise applications unless requirements for low-noise applications are specified (see 6 2 1)

4 7 14 Dielectric strength and insulation resistance tests

4 7 14 1 Dielectric strength. Dielectric strength between electrical circuits and between electrical circuits and ground shall be tested in accordance with MIL-E-16400 For first article tests, the qualification inspection conditions of MIL-E-16400 shall apply, otherwise the acceptance inspection conditions shall apply In the case of low voltage components, waivers shall be obtained in specific instances permitting dielectric tests to be made at less than 600 volts in consideration of reasons of compelling importance Gyro assemblies shall not be given this test

4 7 14 2 Insulation resistance Immediately after passing the dielectric test each circuit of each instrument or unit shall be tested for insulation resistance in accordance with MIL-E-16400 A potential of 500 Vdc shall be applied between a terminal and the ground when the temperature of the instrument is approximately the normal operating temperature

4 7 15 Effect of magnetic field The compass unit with its binnacle shall be given a test to determine the effect of degaussing The effect of magnetic fields on the compass unit shall be determined by using Helmholtz coils that shall produce a magnetic field of 10 gausses uniform intensity and direction over a cube 3 feet on a side when the permeability of the cube is 1 No adjustment of the field strength shall be made after insertion of the equipment between the coils The compass shall be mounted midway between the coils with the gyro rotor approximately on the axis of the coils, and shall be settled with no field applied The field shall be applied in any direction, the compass unit allowed to settle, and the maximum error and new settling position noted In no case shall be settling with the field applied exceed 4 hours. The direction of the magnetic field shall then be reversed and the procedure repeated These two

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tests shall be repeated for each of the two directions perpendicular to the original direction of the magnetic field and for a direction of the magnetic field which shall be inclined 45 degrees in the vertical plane parallel to the gyro spin axis The maximum compass error shall not exceed the value specified in 3 5 10

4 7 16 Shock and vibration One of each component of the equipment shall be subjected to the shock and vibration tests to determine conformance to 3 5 and 3 5 The compass unit shall be settled as defined in 6 4 6 before the test During the vibration test the compass unit shall not exceed an error of plus or minus 0 75 degree During the shock test the compass unit shall not exceed an error of plus or minus 4 degrees After both the shock and vibration tests the compass unit shall settle with an error not in excess of the settling error allowed in 3 5 10 All other units of the compass equipment shall, to meet the shock and vibration tests, not suffer a failure which would interfere with proper compass operation during or after the test

4 7 17 Maximum rate of turn The compass unit shall be settled as defined in 6 4 6 on any heading designated by the Government representative The turntable shall be turned in azimuth in the following manner The platform shall be smoothly accelerated from rest to a velocity of 100 degrees per second, rotated at a constant velocity of 100 degrees per second for approximately one revolution and then smoothly decelerated to rest During the constant velocity part of the motion of the overall lag shall be determined by energizing from the gyro compass, a control transformer driven by the turntable and measuring the output voltage The actual lag shall be computed from the control transformer calibration and shall not exceed 15 minutes of arc The compass unit error as determined from the output of the calibrated control transformer at 5-minute intervals for 30 minutes, beginning 5 minutes after the rotation, shall not exceed 1 degree

4 7 18 Angular freedom The test for angular freedom shall be performed as follows Settle the compass as defined in 6 4 6 on one heading Thereupon tilt the turntable 45 degrees to port in roll and maintain the turntable in this attitude approximately 30 minutes, then repeat for a 45-degree roll to starboard Repeat this procedure three times Thereafter do the same procedure for a 45-degree downward pitch and a 45-degree upward pitch The roll and pitch attitudes should be obtained at slow rates, approximately 10 degrees per second At no time during the test should the compass become unsettled

4 7 19 Test of alarm system The alarm system shall be tested to determine conformance with 3 5 4

4 8 Spare sensitive element When a spare gyroscopic element assembly is provided in the contract, it shall be tested in any compass unit of that same contract selected by the contractor with concurrence of the Government representative The inspection shall consist of the following tests and shall be conducted under the same conditions as for the compass unit.

(a) Rotor vibration (see 4 7.12)
(b) Damping (see 4 7 4).
(c) Turntable (see 4 7 6)

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(d) Small angle Scorsby (see 4.7 7)
(e) Insulation (see 4 7 14)

4 9 Failure The occurrence of any of the following compass unit conditions shall require the subsequent running of the turntable test, small angle Scorsby test, large angle Scorsby test and such other tests as the contractor with concurrence of cognizant Government representative may consider to be affected by the failure or the subsequent adjustments

(a) Compass unit error in excess of the allowable limits during the turntable test, small angle Scorsby test, large angle Scorsby test, except as provided in 4 9 2
(b) Failure to settle as defined in 6 4 6 or failure to remain settled at any time during the test
(c) Failure, defective operation, breakdown on the insulation test, or need for adjustment of any part of the gyroscopic element or follow-up system, including amplifier, at any time during the test
(d) Loss of rotor case vacuum

4 9 1 Other failure Except as provided in 4 9, any failure of the compass unit to conform to the specified accuracy or other limit of operation or failure of any auxiliary component of the equipment affecting the operation of the compass unit shall require the subsequent running of those tests which the contractor with concurrence of cognizant Government representative may consider to be affected by the failure of the subsequent adjustments

4 9 2 Trial rerun In case the compass unit error exceeds the allowed limit during the turntable test, small angle Scorsby test or large angle Scorsby test, and there is no apparent cause for this condition, the contractor may request a rerun of those tests on which the excessive error was observed, for the purpose of determining the cause of the error If the reasons for such a run are satisfactory, the rerun shall be made under carefully observed conditions If the results of the rerun are satisfactory, the test in question shall be considered complete and the excessive error during the original run shall not be considered a failure

4 9 3 Auxiliary unit failure A failure of any auxiliary unit of the equipment (except parts of the gyroscopic element and follow-up system) which affects the operation of the compass unit, while being tested as specified for quality assurance, shall require the rerunning of the test on which the failure occurred

4 10 Inspection of packaging Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein

5 PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6 6)

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5 1 Preservation, packing and marking The equipment and accessories shall be preserved level A or C, packed level A or B as specified (see 6 2 1) and marked in accordance with MIL-E-17555 Method II shall apply for level A preservation of equipment

5 2 Service Record Book One Government furnished Service Record Book, NAVSHIPS 708 shall be packed with each gyro compass

6 NOTES

6 1 Intended use The compass is intended for use on board a ship or vehicle to provide the true heading of the carrier

6 2 Ordering data

6 2 1 Acquisition requirements Acquisition documents should specify the following

(a) Title, number and date of this specification
(b) Quantity of items required
(c) Whether first article inspection is required (see 3 1 and 4 3)
(d) Whether alarm unit is for single or multiple compass installation (see 3 2)
(e) If visual bridge alarm indicator is required with each compass (see 3 2)
(f) Whether power supply unit is required with each compass (see 3 2)
(g) Quantity of repair parts (see 4 3 and 6 5)
(h) Grade C equipment for low noise applications, both for airborne audible and structureborne noise (see 3 5 1 4 and 4 7 13)
(i) Level of preservation and packing required (see 5 1)

6 2 2 Data requirements When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) incorporated into the contract When the provisions of DoD FAR Supplement, Part 27 Sub-Part 27 475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements Deliverable data required by this specification are cited in the following paragraphs

Paragraph no.	Data requirement title	Applicable DID no	Option
3 5 12	Drawings, engineering and associated lists	DI-E-7031	---
4 1 2	Inspection system program plan	DI-R-4803	---

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L , AMSDL Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer)

6 2 2 1 The data requirements of 6 2 2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports)

6 2 2 2 Technical manuals The requirement for technical manuals should be considered when this specification is applied on a contract If technical manuals are required, Military specifications and standards which have been cleared and listed in DoD 5010 12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), included as an exhibit to the contract The technical manuals must be acquired under separate contract line item in the contract

6 3 First article When a first article inspection is required, the item should be a first article sample The first article should consist of one unit The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract

6 4 Definitions

6 4 1 Test stand attitude

6 4 2 Test stand heading The test stand heading is defined as the angle between a north and south vertical plane and the vertical plane through the fore and aft line of the Scorsby test plate, measured in the horizontal plane in a clockwise direction from north

6 4 3 Test stand roll The test stand roll is defined as the angle between the plane of the testplate and the horizontal plane, measured in the plane perpendicular to the fore and aft line of the test plate, positive when starboard side of test plate is up

6 4 4 Test stand pitch The test stand pitch is defined as the angle between the plane of the test plate and the horizontal plane, measured in a vertical plane through the fore and aft line of the test plate, positive when the forward end of the test plate is down

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6 4 5 Compass unit error. The compass unit error is defined as the angular difference between the indications of the compass 36-speed transmitter and the test stand attitude The compass error is positive when the compass transmitter indication is greater than the test stand indication

6 4 6 Settling point The settling point of a compass unit on test is defined as the average of three consecutive readings of the compass unit taken at 15-minute intervals during which the compass error at no time exceeds the settling point error specified in 3 5 10 and during which the difference between any two of these readings does not exceed the settling point difference in 3 5 10 For settling point determination the corrective devices are adjusted for zero speed at the local latitude setting The compass is considered to have settled at the start of the first reading from which the settling point is obtained, that is, at the time at which the error becomes (and from which the error subsequently remains) less than the allowed settling point error

6 5 Provisioning Provisioning Technical Documentation (PTD), spare parts and repair parts should be furnished as specified in the contract

6 5 1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment Packaging for such parts should also be specified

6 6 Sub-contracted material and parts The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped

6 7 Subject term (key word) listing

Compass unit
Rate of turn
Roll and pitch
True heading

6 8 Changes from previous issue Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

Preparing activity
Navy - SH
(Project 6605-N388)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-C-15952G(SH)	COMPASS, GYRO, MINIATURE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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